Target Product Profile: PKS1 Monoclonal Antibody Therapy

A product profile provides a structure for the scientific, technical, clinical, and market information that is required to achieve a desired commercial outcome. It provides all stakeholders with a clear vision of the product objectives and helps guide research and development decisions. It is a dynamic strategic document that should be reviewed throughout the development process. *It identifies the aspirational and minimal product attributes and claims that are key indicators for success. These will be the thresholds that you will use to determine if your candidate's testing and metrics are acceptable.*

Last Updated: 04/03/2024

Product Name	
Product Description Brief description of the therapeutic including its purpose, route of administration, dosage, and if it will be accompanied by a companion diagnostic.	Novel monoclonal antibody for PKS1 (PKS1AB) to reduce Parkinson's Disease (PD) symptoms.
Unmet Clinical Need Description of the unmet clinical need and how the product will change medical practice. Consider how the clinical need is currently addressed.	To date, there is no cure for PD and the current standard of care focuses on treating symptoms rather than the disease its self. This monoclonal antibody is the first therapy targeting PKS1 to slow or halt the progression of PD.
Indication Target disease or manifestation of a disease and/or population. Include examples of specific patient population(s).	The target population are adults 18 and older who have a diagnosis of PD and elevated blood plasma levels of PKS1.
Comparative Therapeutic Current therapeutic(s) that will be used for comparison and a brief competitive analysis of developing therapeutics	Prasinezumab is a competitor monoclonal antibody treatment currently being developed to slow the progression of motor symptoms in PD. It works by targeting the aggregated alpha-synuclein protein which is attributed to progression of PD. What is different about our product, is that it is the only therapeutic targeting the PKS1 pathway.
Mechanism of Action (MOA) The mechanism by which the product produces an effect on a living organism.	The mechanism of action is a monoclonal antibody targeting the PKS1 gene to downregulate expression of the PKS1 protein.

Commented [JP1]: Do we need this section since there is no existing therapy comparable to our mAb?

Commented [MM1R2]: perhaps we could compare it to other mAbs on the market or compare it to the therapeutic that was in the IP that was studied: Sinemet

Commented [MM1R3]: If you want to compare it to sinement, there is a table of ppl with PD taking the drug and their PKS1 levels.

Clinical Pharmacology Pharmacokinetic information, distribution and pathways for transformation. *This should **NOT** be the specific pharmacology of the compound you are investigating. This information should list the acceptable criteria a developmental candidate must meet to be considered a lead candidate. The criteria will guide the lead clinical candidate decision making process and provide you with a checklist of metrics your compound must meet.*	This will be determined once preclinical trials have been conducted.
Target Metrics List your target and include the following metrics: ***Small molecule targets***: druggability, novelty, validation, and potential mechanistic-based toxicity ***Antibody targets***: if it is a surface protein, likelihood of delivering a therapeutic effect with your specific antibody approach (naked antibody, conjugated antibody, immunotherapy), validation, and specificity of expression.	This will be determined once preclinical trials have been conducted.

Commented [JP2]: I'm not super familiar with mAbs. Would like to ask the Prof where I can find information to help me fill in last two sections

Developmental Candidate Selection Criteria	
Freedom to Operate and IP Position Consider the novelty of your technology and patent eligibility. Describe the extent of interactions with the technology transfer office. Please include a list of the various types of IP filed or granted (including patent application number, issued patent number, title, status and date) and the major types of claims (e.g., composition of matter claims, use claims, process of manufacturing claims). Also list the IP events anticipated during development.	Patent Application Number: 8,675,309 Issued Patent Number: 12/116,876 Title: PKS1 protein biomarker and use thereof Status and Date: (Date Filed) Jun 22, 2012 Major types of claims: 1. A method for the treatment of Parkinson's disease comprising of administering a therapeutically effective amount of an agent that interacts with or modulates the expression or activity of a PKS1 protein.
Manufacturability The ease and cost of manufacturing including: availability and cost of starting material; scalability of manufacturing process; total yield from readily available starting material; and competitor's cost.	**Availability and cost of starting Material:** Will be determined later **Scalability of the manufacturing process:** Currently we have an immortal hybridoma cell line that produces the mAb. This cell line is able to be scaled up. Eventually can use CHO cells to produce the mAb through bioreactors and purify through chromatography. **Total yield from readily available starting material:** Will be determined later. **Competitor's cost:** Will be determined later. **Helpful Links:** **FDA guidance with Hybridoma:** Guidance for Industry: Monoclonal Antibodies Used as Reagents in Drug Manufacturing (fda.gov) **Other FDA guidance:** Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use | FDA **Cost to manufacture a new drug:** Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009-2018 - PubMed (nih.gov) Pricing of monoclonal antibody therapies: higher if used for cancer - PubMed (nih.gov) **Competitors mABs:** Monoclonal Antibodies as Neurological Therapeutics - PMC (nih.gov)

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<td><u>Developmental Candidate Potency</u>

The developmental candidate’s targeted potency in a biochemical assay, cell-based assay and animal model. Consider potency of competitors.

This should <u>NOT</u> be the observed measurements of the compound you are investigating. This information should list the acceptable criteria a developmental candidate must meet to be considered a lead candidate. The criteria will guide the lead clinical candidate decision making process and provide you with a checklist of metrics your compound must meet.</td>
<td>Biochemical assays:
Titer assay to measure which is the lead candidate of producing the most amount of mAb.

Quality assays: SDS nonreduced and reduced, cIEF, glycosylation

Binding affinity and interactions with the mAb with other structures that are closely related to the intended target.

Amino Acid sequencing.</td>
<td>Cell-based assay:
Flow cytometry to see total RNA content, total protein content, cell cycle phase distribution, concentration of light and heavy chains concentration

Helpful Link:
<u>Comparison of the production of a human monoclonal antibody against HIV-1 by heterohybridoma cells and recombinant CHO cells: A flow cytometric study - PubMed (nih.gov)</u></td>
<td>Animal model:
Reduce symptoms in both mice models for PD:
JAX®mice strain: 006582 and 009090</td>
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<td><u>Developmental Candidate Specificity</u>

The developmental candidate’s targeted specificity as compared to other members of the target family, and against a broader panel of proteins.

This should <u>NOT</u> be the observed measurements of the compound you are investigating. This information should list the acceptable criteria a developmental candidate must meet to be considered a lead candidate. The criteria will guide the lead clinical candidate decision making process and provide you with a checklist of metrics your compound must meet.</td>
<td colspan="2">Compared to other members of the target family:
Sinemet: a combination drug dopamine replacement therapy which is given to treat the symptoms of PD. It is used to preserve normal dopamine levels
Consists of
1. Levodopa which is then converted to dopamine within the brain.
2. Cardidopa which enhances levodopa, and allows for a lower dose to be given.
Our product: a mAb that can interfere with the transcription of the PKS1 protein.</td>
<td>Against a broader panel of proteins:

α-Synuclein: a protein that regulates synaptic vesicle trafficking and neurotransmitter release. Is linked genetically and neuropathologically to PD. Abundant in the brain and smaller amounts in heart, muscles, and other tissues. When excreted, can have an effect on intracellular targets such as synaptic function.

Helpful links:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3281589/</td>
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<td><u>Biocompatibility</u>

ADME/PK Properties, on-off rate, solubility, stability

This should <u>NOT</u> be the observed measurements of the compound you are investigating. This information should list the target and minimal criteria a developmental candidate must meet to be considered a lead candidate. The criteria will guide the lead clinical candidate decision making process and provide you with a checklist of metrics your compound must meet.</td>
<td>Target:

Will be evaluated at a later date</td>
<td>Minimal Condition:

Will be evaluated at a later date</td>
<td>Competitor/Optimal Conditions:

Will be evaluated at a later date.

(Perhaps Sinemet: Carbidopa and Levodopa)</td>
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Commented [MM3]: https://www.drugdiscoverynews.com/a-guide-for-potency-assay-development-of-cell-based-product-candidates-7802

Commented [MM3R2]: Potency assay will represent the product's known or intended MoA.

Commented [MM4]: Compare to other mAbs that are used for neurological disease like Ocrelizumab for MS or others for migranes

Commented [MM4R2]: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7912592/ Table 1

Safety & Toxicology	***Target:***	***Minimal:***	**Competitor**:
Safety & toxicology profile and requirements to be met (e.g. non-clinical toxicology). Include testing that will be conducted prior to entering clinical testing. *This should **NOT** be the observed measurements of the compound you are investigating. This information should list the target and minimal criteria a developmental candidate must meet to be considered a lead candidate. The criteria will guide the lead clinical candidate decision making process and provide you with a checklist of metrics your compound must meet.*	-Adequate Acceptable Daily Exposure calculations of the therapeutic. -understanding off target binding with other types of receptors, ion channels, transporters, enzymes, and other targets which can cause adverse effects. Determine future tox studies	Testing for Endotoxin and Bioburden -the maximum tolerated dose -find out what other organs toxicity is seen. Basic principles of non-clinical development: Toxicology (eupati.eu)	Risk for mAb as a whole is their inherent risk for adverse immune-mediated drug reactions such as infusion reactions, cytokine storm, immunosuppression and autoimmunity.

Commented [MM5]: https://link.springer.com/chapter/10.1007/164_2015_16

Lead Candidate Criteria			
Primary Efficacy Endpoints The most important clinical outcome measure. Ideally should be easy to interpret and sensitive to treatment differences. *This information should list the target and minimal efficacy endpoints the clinical candidate must meet to obtain a successful positive clinical trial result.*	***Target:*** The target measure that the therapeutic solution should achieve is a level of PKS1 downregulation such that the risk of Parkinson's Disease onset is significantly reduced.	***Minimal:*** Ensure the therapeutic agent is functional and can downregulate PKS1 expression to some extent.	***Competitor/Optimal Conditions:*** Optimal treatment measures: - Ideal outcome is to eliminate risk of PD in patients with high risk (i.e. history in family, unusually high levels of PKS1 early on) Current PD treatment: - Carbidopa and Levodopa, intended to treat symptoms of PD onset
Secondary Efficacy Endpoints Additional criteria that may be met during a clinical trial, but that are not required to obtain a successful positive clinical trial result	***Target:*** Continuation of therapy should eliminate PD symptoms, if observed in patients.	***Minimal:*** If PD onset has occurred and patient exhibits PD symptoms, long term patient adherence should lower symptom frequency to a considerable extent.	***Competitor/Optimal Conditions:*** Current PD treatment focuses on symptoms (carbidopa-levodopa). The idea for this therapeutic is sort of an all-in-one: target PD before onset, and if onset has occurred, additionally target the symptoms to reduce their frequency.
Dosage and Administration Route and site of administration, dosage volume and frequency, administrating location and personnel. *This information should list the target and minimal criteria the clinical candidate must meet.*	***Target:*** Biologic: - Perispinal injection solution, to give therapeutic easier access across blood-brain barrier - Second option: subcutaneous injection like insulin (i.e. vials or injectable pens/autoinjectors). Subcutaneous entry will allow therapeutic to enter blood flow.	***Minimal:*** Administration, if therapeutic is a biologic product, will be via perispinal injection. Could be given at a medical office during routine checkups for early PD evaluation and treatment	***Competitor/Optimal Conditions:*** Optimal conditions: - Method of administration would be at-home injection solutions (i.e. like insulin pens/vials), where effects of therapeutic would be measured and evaluated by IVD to determine treatment for patient moving forward.

Commented [SN6]: I'm not too sure on this section. If the therapeutic intention is to downregulate/inhibit PKS1, would this be referring to a potential side effect or an off-label use? If so, I'm not entirely sure on what exactly to put down here.

Commented [MM6R2]: The FDA has a guidance on this, https://www.fda.gov/media/162416/download It is in section three. It gives an example of "an effect on survival when a cardiovascular drug has show an effect on the primary endpoint of heart failure related hospitalizations" perhaps maybe targeting an increase of quality of life ?

Commented [SN6R3]: Makes sense. I'll think on this as I work on this section more, as I'm still trying to think on optimal/minimal conditions for the primary efficacy point.

Commented [MM7]: are we still thinking of giving an perispinal injection vs subcutaneous injection try to better get past the BBB?

<u>Contraindications</u> Known or expected contraindications	**Common contraindications for biologics:** - Possibility of infection. Biologics and related biosimilars are derived from living organisms, so the therapeutic's entry into the blood flow can pose a risk of infection and can trigger unfavorable responses from the patient's immune system.
<u>Packaging & Labeling</u> Packaging and labeling necessary for therapeutic. Consider how it will be stored and administered; home or hospital setting; the expected shelf life, labeling of measurements etc. *Packaging and labeling of comparative therapeutics should be referenced.*	**Biologic:** - Vial or injectable pen stored in box, similarly to insulin biologics - General labeling standards should be followed as per 21 C.F.R 201. Packaging should contain manufacturing information, national drug code information, ingredients (active/inactive), expiration date, and general directions for use. - Insulin and comparable biologics for either diabetics or weight loss are kept refrigerated due to the active ingredient being less effective at room temperatures. o https://www.fda.gov/drugs/emergency-preparedness-drugs/information-regarding-insulin-storage-and-switching-between-products-emergency#:~:text=The%20longer%20the%20exposure%20to,insulin%20as%20cool%20as%20possible.
<u>Preclinical Testing</u> Additional testing (e.g. efficacy) that will be performed prior to entering clinical testing, including bench, in vivo, computer simulations, sterility, biocompatibility, robustness, etc. Consider how you will scale-up for testing.	**General preclinical testing for biologics:** - Toxicology evaluation: repeat-dose toxicity testing to determine presence of immunogenicity (ability for cells to provoke an unfavorable physiological response from the immune system) that could have detrimental effects on exposure to long-term dosing - Completion of Investigational New Drug (IND) application, provided with results of preclinical testing. - Testing will allow researchers to evaluate immunogenic potential of developmental drug (in our case, PKS1 regulation therapeutic). - https://www.pharmamodels.net/blog/toxicology-testing-biologics-pre-clinical-studies/#:~:text=In%20the%20preclinical%20stage%2C%20conventional,potential%20immunogenic%20epitopes%20in%20biologics. - https://www.fda.gov/drugs/investigational-new-drug-ind-application/drug-development-and-review-definitions#:~:text=At%20the%20preclinical%20stage%2C%20the,months%2C%20depending%20on%20the%20proposed

Commented [SN8]: General FDA guidelines for preclinical drug development and requirements

Clinical Studies Study model, testing, and clinical data necessary to support use and for FDA clearance. Consider competitive position and marketing strategy.	https://www.fda.gov/drugs/therapeutic-biologics-applications-bla/frequently-asked-questions-about-therapeutic-biological-products With successful completion and approval of an Investigational New Drug (IND) application, clinical testing and trials in humans can proceed. Testing will evaluate if the biologic is reasonably safe and effective for intended use. A biologics license application (BLA) would also need to be completed alongside the New Drug Application (NDA) for market approval, under the Public Health Service Act.